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Resource Management Training as a Non-Pharmacological Resilience Intervention Effects on Salivary Cortisol and Emergency Decision-Making in Trauma-Exposed Adults

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Abstract

- Acute psychosocial stress can impair prioritisation, situational awareness and error monitoring in safety-critical work.
- This proposal tests whether checklist-embedded resource management training reduces salivary cortisol reactivity and supports emergency decision-making in trauma-exposed adults.
- The predicted effect is lower cortisol and fewer procedural errors, especially at higher perceived stress/PTSD symptom levels.

Introduction

- Acute stress weakens prefrontal functions needed for flexible control and error monitoring (Arnsten, 2009).
- In safety-critical systems, goals and routines can drift toward unsafe boundaries under high workload (Dekker, 2011).
- Resource management training strengthens situational awareness, communication and shared verification under pressure (Buljac-Samardžić et al., 2021).
- Salivary cortisol is a practical, non-invasive index of hypothalamic-pituitary-adrenal (HPA) axis reactivity and recovery (Hellhammer et al., 2009).

Research Question & Hypotheses

Does resource management training attenuate salivary cortisol reactivity and improve emergency decision-making after acute psychosocial stress, and are effects moderated by perceived stress and PTSD symptoms?

H1: Stress will increase cortisol and impair performance.
H2: Training will reduce errors and attenuate cortisol reactivity under stress.
H3: Higher baseline stress/PTSD symptoms will predict larger stress responses and a larger training benefit.

Research Design & Method

2 × 2 Between-Subjects Factorial RCT

	e-TSST stress	Neutral control
Resource management	RM + stress	RM + neutral
Active control	Control + stress	Control + neutral

Sample & Materials

N ≈ 80 cruise ship officers, aged 20-50.
Recruitment: maritime academy and training networks.
Resource management training (45 min): briefing, think-aloud communication, cross-checks and checklist-based debriefing.
Active control: matched for duration and format, but without decision prompts.

Procedure & Measures

Consent + baseline questionnaires, then training/control and e-TSST stress or neutral condition (Hawn et al., 2015; Kirschbaum et al., 1993).
Saliva: T0 after a 10-minute rest, then +20, +35 and +45 minutes.
Emergency simulator task follows T1.
Outcomes: cortisol, procedural errors, decision quality and response time.
Moderators: perceived stress and PTSD symptoms.

Analysis & Ethics

Linear mixed-effects models test Training, Condition and their interaction. Screening, afternoon scheduling, monitoring and structured debriefing reduce risk and measurement noise.

Limitations

- Simulator tasks may not fully capture real-world trauma.
- Volunteer recruitment may introduce self-selection bias.
- Cortisol is sensitive to time of day, medication, nicotine and oral health.
- PTSD symptoms are self-reported, so inference concerns symptom severity, not diagnosis.

Take-home message: Structured briefing, cross-checks and checklist-based debriefing may offer a practical, non-pharmacological route to resilience under acute stress.

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